

HUSSAR AGDM, AB, Canada

WMO: 713440

Lat: 51.1911N

Lon: 112.5027W

Elev: 3186

StdP: 13.08

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-18.7	-13.7	-25.3	1.5	-18.6	-19.3	2.1	-13.4	31.5	24.8	28.2	21.9	9.7	320	0.746

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	83.7	61.3	80.1	60.1	76.6	59.0	65.2	77.0	63.0	75.0	61.1	73.0	9.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.8	89.6	71.9	58.4	82.0	68.5	56.3	75.9	66.2	31.9	77.2	30.1	75.2	28.7	73.1	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.5	24.9	21.9	DB	-25.4	89.8	4.6	3.7	-28.7	92.4	-31.4	94.6	-33.9	96.7	-37.3	99.4	(4)
(5)			WB	-25.6	70.6	4.4	2.5	-28.7	72.4	-31.3	73.8	-33.8	75.2	-37.0	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.1	17.9	16.0	26.7	39.0	50.2	57.0	62.5	60.7	52.6	40.6	26.7	17.7
	DBStd	20.11	15.59	14.63	14.06	9.61	8.46	5.80	5.63	6.49	8.92	10.01	12.96	14.01
	HDD50	5220	996	953	723	342	102	8	0	5	73	317	701	1000
	HDD65	9538	1461	1373	1187	779	460	247	115	163	382	756	1150	1465
	CDD50	1245	0	0	1	13	110	219	389	336	150	26	1	0
	CDD65	88	0	0	0	0	3	8	39	29	9	0	0	0
	CDH74	1346	0	0	0	4	74	121	533	447	161	6	0	0
	CDH80	321	0	0	0	0	9	15	148	113	36	0	0	0
Wind	WSAvg	11.2	12.7	11.7	11.7	11.9	12.1	11.3	9.1	9.1	10.4	11.3	11.8	11.7
Precipitation	PrecAvg	15.1	0.6	0.4	0.6	1.0	1.9	3.0	2.6	2.1	1.3	0.6	0.6	0.5
	PrecMax	19.3	1.3	0.9	2.1	2.4	3.7	6.0	5.9	4.3	3.2	1.4	1.9	1.1
	PrecMin	9.0	0.0	0.0	0.1	0.1	0.2	1.3	0.9	0.1	0.2	0.2	0.1	0.0
	PrecStd	2.6	0.3	0.2	0.4	0.6	0.9	1.2	1.2	1.1	0.7	0.4	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.7	47.0	59.3	71.0	80.7	81.9	89.2	87.6	84.6	74.0	58.7	48.0
		MCWB	37.4	37.0	45.0	49.9	54.4	61.7	63.6	59.6	57.9	52.4	45.9	38.3
	2%	DB	41.9	41.4	52.3	65.3	75.5	77.4	84.1	83.0	79.1	65.5	51.8	43.0
		MCWB	35.3	34.7	40.9	47.0	53.3	58.7	63.9	60.1	56.3	49.1	41.3	36.4
	5%	DB	39.7	37.7	47.5	59.6	70.8	73.5	79.8	79.1	73.5	60.0	46.6	39.5
		MCWB	34.3	32.3	38.4	43.8	51.1	57.1	63.2	59.8	54.4	46.4	38.7	33.8
	10%	DB	37.2	34.3	42.8	54.6	66.3	69.9	75.9	75.0	68.2	54.9	42.3	36.1
		MCWB	32.7	30.0	35.9	41.1	48.9	55.6	61.7	59.0	52.3	44.1	35.7	31.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.5	38.2	46.0	50.8	56.6	65.1	70.3	65.7	59.7	53.2	45.8	39.6
		MCDB	44.6	44.8	57.1	69.6	76.0	77.0	81.8	76.6	77.7	71.2	58.1	45.5
	2%	WB	36.0	35.0	41.9	47.5	54.1	61.5	66.7	63.4	57.3	50.2	41.9	36.7
		MCDB	40.9	40.9	50.9	62.5	71.0	72.8	78.4	76.1	74.6	62.2	50.4	42.6
	5%	WB	34.6	32.6	38.9	44.8	52.3	58.8	64.4	61.6	55.2	47.3	39.2	33.8
		MCDB	39.4	37.2	46.1	57.6	67.7	69.7	76.2	74.6	71.4	59.0	45.7	39.3
	10%	WB	32.6	30.3	36.2	42.2	50.4	56.8	62.4	59.7	53.1	44.7	36.3	31.2
		MCDB	37.2	33.9	42.3	52.8	63.7	67.4	74.0	72.4	66.8	54.1	41.5	35.9
Mean Daily Temperature Range	5% DB	MDBR	15.5	15.5	16.4	20.0	21.5	20.1	22.9	22.8	21.8	19.4	15.5	14.6
		MCDBR	15.8	18.0	22.5	28.8	27.9	25.8	28.6	29.9	29.8	26.2	20.6	16.1
		MCWBR	11.6	13.1	13.7	14.5	12.2	12.1	14.3	12.6	12.8	13.6	12.9	11.5
	5% WB	MCDBR	15.8	17.4	21.5	26.6	24.9	22.7	25.5	25.7	27.5	25.3	20.5	16.5
MCWBR		12.1	13.0	13.8	14.2	11.5	12.3	14.5	12.6	12.6	13.6	13.6	12.4	
Clear-Sky Solar Irradiance	taub	0.260	0.268	0.280	0.334	0.359	0.351	0.349	0.366	0.320	0.279	0.250	0.239	
	taud	2.381	2.435	2.479	2.335	2.311	2.382	2.416	2.345	2.449	2.545	2.498	2.407	
	Ebn at Noon	248	279	296	287	284	286	285	274	277	273	255	240	
	Edh at Noon	19	23	27	36	38	36	35	35	28	21	16	16	
All-Sky Solar Radiation	RadAvg	360	681	1112	1513	1823	1966	2112	1703	1201	730	392	279	
	RadStd	27	59	91	115	123	120	104	117	121	85	33	27	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (81 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page